QChain — One-Page Summary

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QChain

A research blockchain combining post-quantum signatures (CRYSTALS-Dilithium), quantum-randomness-driven leader election (IBM Quantum Network QRNG), and a hand-rolled zk-STARK system (Goldilocks AIR via Winterfell) for shielded payments.

What state is it in

Metric	Value
Total tests (Rust + PyO3 + Python)	443
Maintained Markdown documents	37
Numbered threats with explicit status	23
Threats `[DEFENDED]` or stronger	20
Threats `[NOT DEFENDED]` (documented)	2
Self-found bugs/defects, all disclosed	5

All tests green. Code: https://github.com/sil714/qchain

What's interesting about it

The cryptographic primitives are well-known. What's unusual is the **engineering discipline applied to documenting and testing the system**:

- Every defended attack carries explicit code references and tests.
- Every limitation is a numbered threat with one of five status markers: `[FORMAL]`, `[FORMAL, MODULO]`, `[DEFENDED]`, `[HEURISTIC]`, `[NOT DEFENDED]` — distinguishing "defended under a cryptographic assumption" from "defended by application-layer mechanism" from "explicitly out of scope."
- Five bugs found and fixed by the project's own audit-readiness work, including a coinbase-inflation flaw, a transparent-tx replay bug (caught by Hypothesis property testing), and an infinite-loop protocol bug surfaced when rate limiting was added.
- Differential testing cross-checks the zk-STARK trace against an independent Python re-execution of Rescue-Prime, targeting the "circuit is wrong but positive examples still work" failure mode.

What it is not

A production blockchain. No BFT proof-of-stake. No peer authentication or TLS. No formal proof of AIR correctness (differential testing is the project's best evidence, not a proof). No peer-reviewed cryptographic construction. The project is a **research demo** built with the discipline of something more serious, and made deliberately legible to someone considering external engagement.

What would help most

An independent review of the STARK circuits for shielded payments by a formal-methods specialist or cryptographer. Several threats marked `[FORMAL, MODULO]` ride on AIR correctness as their non-formal assumption — the highest-leverage external contribution.

Read more

`PUBLICATION.md / .pdf` (13-page retrospective) · `AUDIT-PACKAGE.md` (claims C1–C10) · `THREAT-MODEL.md` (23-threat status table) · `DOCS.md` (reading order)